A3 Delta Lake

MERGE, idempotent writes, schema enforcement and evolution, time travel, maintenance.

A3-L1 Idempotent MERGE, and why you deduplicate the source first

The strong version is idempotent for any source: repeated keys, a retry or a full redelivery. The weak version works only while the supplier never repeats a row.

When is a pipeline idempotent?	When running it twice with the same input leaves the table as one run would.
Two conditions for an idempotent Delta MERGE	The ON condition covers the full grain, and the source has at most one row per key.
What happens when two source rows match one target row?	The MERGE fails with DELTA_MULTIPLE_SOURCE_ROW_MATCHING_TARGET_ROW_IN_MERGE.
Why not dropDuplicates() before a MERGE?	It keeps an arbitrary row per key. Use row_number() over a window ordered by version, newest first.

A3-L2 Schema enforcement is on by default. Evolve on purpose.

Schema changes become decisions a person reviews, not accidents a delivery causes.

Delta append with an extra column, no options	Fails with a schema mismatch. Enforcement is on by default.
MERGE with updateAll and an extra source column, no evolution	Succeeds and silently ignores the extra column. Only your own column check catches it.
Evolve one write on purpose	Append: option mergeSchema. MERGE: .withSchemaEvolution(). Planned: ALTER TABLE ADD COLUMN.
Why not set autoMerge.enabled for the session?	It turns evolution on for every MERGE in the job, including ones you did not intend.

A3-L3 Time travel, RESTORE, OPTIMIZE and VACUUM

History turns a data incident into a two-minute restore. Keep it long enough to be useful.

Read an old Delta version	VERSION AS OF n or TIMESTAMP AS OF ts in SQL; option versionAsOf in PySpark.
What does RESTORE do to history?	Adds a new version with the old content. Nothing is deleted.
Default retention windows	Deleted files 7 days (delta.deletedFileRetentionDuration). Log 30 days (delta.logRetentionDuration).
Why is VACUUM ... RETAIN 0 HOURS dangerous?	It deletes the files that time travel and RESTORE need, and it breaks readers still using them.

Databricks' recommendation for layout and maintenance on UC managed tables	Liquid clustering (CLUSTER BY or CLUSTER BY AUTO) with predictive optimization.